

# Task 4: Cross-Functional SQL Analysis & Results

## Summary

**Project:** Gen P Technologies - Cross-Functional Capstone  
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This document details the relational database modeling and SQL scripting performed to integrate siloed department data (Sales, Marketing, Finance, and Operations) into unified executive insights.

### 1. Marketing Analysis

Bridging Marketing campaign data with closed-won sales revenue and lifetime financial metrics to evaluate true channel profitability.

#### Unit Economics (LTV:CAC Ratio) by Channel

```
WITH Channel_CAC AS (  
    SELECT  
        m.Campaign_Type AS Acquisition_Channel,  
        SUM(m.Actual_Spend_USD) / NULLIF(SUM(CASE WHEN s.Sales_Stage = 'Closed Won' THEN 1 ELSE 0  
END), 0) AS CAC  
    FROM marketing_campaigns m  
    JOIN `sales_pipeline cleaned data` s ON m.Campaign_ID = s.Campaign_ID  
    GROUP BY m.Campaign_Type  
)  
Channel_LTV AS (  
    SELECT  
        Acquisition_Channel,  
        AVG(LTV_USD) AS Average_LTV  
    FROM saas_financials  
    GROUP BY Acquisition_Channel  
)  
SELECT  
    c.Acquisition_Channel,  
    ROUND(c.CAC, 2) AS Customer_Acquisition_Cost,  
    ROUND(l.Average_LTV, 2) AS Average_Lifetime_Value,  
    ROUND(l.Average_LTV / NULLIF(c.CAC, 0), 2) AS LTV_to_CAC_Ratio  
FROM Channel_CAC c  
JOIN Channel_LTV l ON c.Acquisition_Channel = l.Acquisition_Channel  
ORDER BY LTV_to_CAC_Ratio DESC;
```

**Executive Insight:** The analysis identified critical inefficiencies in current marketing spend. Channels such as Paid Search, Webinars, and LinkedIn Ads are operating with an LTV to CAC ratio below 1.0 (e.g., 0.58 to 0.60). This indicates these acquisition channels are currently unprofitable and require immediate strategic optimization.

### 2. Operations Analysis

Evaluating customer support performance, SLA adherence, and the impact of automation on customer satisfaction.

#### SLA Breach & Critical Ticket Escalations

```
SELECT  
    s.Customer_ID,
```

```

        c.Renewal_Status,
        COUNT(s.Ticket_ID) AS High_Critical_Ticket_Count
FROM support_tickets s
JOIN customers c ON s.Customer_ID = c.Customer_ID
WHERE s.Priority IN ('High', 'Critical')
GROUP BY s.Customer_ID, c.Renewal_Status
HAVING COUNT(s.Ticket_ID) > 5
ORDER BY High_Critical_Ticket_Count DESC;

```

### 3. Cross-Functional Analysis

Complex multi-table queries integrating CSAT/NPS, churn rates, and overall regional health into a unified management scorecard.

#### The Churn & Satisfaction Link

```

SELECT
    c.Customer_Tier,
    CASE
        WHEN c.NPS_Score >= 9 THEN 'promoter (9-10)'
        WHEN c.NPS_Score >= 7 THEN 'passive (7-8)'
        ELSE 'detractor (0-6)'
    END AS nps_score_category,
    ROUND(AVG(c.NPS_Score)) AS avg_nps_score,
    COUNT(DISTINCT s.Ticket_ID) AS total_tickets,
    ROUND(AVG(f.Monthly_Churn_Rate_Pct)) AS avg_churnrate
FROM customers c
JOIN saas_financials f ON c.Customer_ID = f.Customer_ID
LEFT JOIN support_tickets s ON c.Customer_ID = s.Customer_ID
GROUP BY c.Customer_Tier, nps_score_category
ORDER BY avg_nps_score DESC;

```

#### Ultimate Executive Region Scorecard

```

WITH Region_Sales AS (
    SELECT Region, SUM(Deal_Value_USD) AS Total_ARR
    FROM `sales_pipeline cleaned data`
    WHERE Sales_Stage = 'Closed Won' GROUP BY Region
),
Region_Marketing AS (
    SELECT s.Region, AVG(m.Activation_Rate_Pct) AS Avg_Activation
    FROM marketing_campaigns m
    JOIN `sales_pipeline cleaned data` s ON m.Campaign_ID = s.Campaign_ID GROUP BY s.Region
),
Region_Support AS (
    SELECT s.Region,
        SUM(CASE WHEN t.Resolution_Hours > t.SLA_Target_Hours THEN 1 ELSE 0 END) * 100.0 /
        COUNT(t.Ticket_ID) AS SLA_Breach_Rate
    FROM support_tickets t
    JOIN `sales_pipeline cleaned data` s ON t.Customer_ID = s.Customer_ID GROUP BY s.Region
),
Region_Finance AS (
    SELECT s.Region, AVG(f.LTV_USD) AS Average_LTV
    FROM saas_financials f
    JOIN `sales_pipeline cleaned data` s ON f.Customer_ID = s.Customer_ID GROUP BY s.Region
)
SELECT
    rs.Region, rs.Total_ARR, ROUND(rm.Avg_Activation, 2) AS Marketing_Activation_Rate,
    ROUND(rsupp.SLA_Breach_Rate, 2) AS Support_SLA_Breach_Rate, ROUND(rf.Average_LTV, 2) AS

```

```
Finance_Average_LTV
FROM Region_Sales rs
JOIN Region_Marketing rm ON rs.Region = rm.Region
JOIN Region_Support rsupp ON rs.Region = rsupp.Region
JOIN Region_Finance rf ON rs.Region = rf.Region
ORDER BY rs.Total_ARR DESC;
```